

Frontend Architecture & Execution

ShopEZ E-commerce Application

1. Frontend Structure

The frontend of ShopEZ acts as the presentation layer and is built using a **component-based architecture**. This means the UI is divided into reusable pieces, which improves maintainability and allows new features to be integrated without breaking existing functionality.

User Interfaces

- **Customer View:** Includes components for creating accounts, secure login, browsing products by category, searching, viewing product details, cart management, wishlist, profile updates, and order tracking.
- **Admin Dashboard:** A dedicated interface for administrators to manage products, categories, users, inventory, and customer orders.

Core Mechanisms

- **State Management:** Used to seamlessly maintain application data across sessions, such as logged-in user information, shopping cart contents, selected products, and authentication status.
- **Client-Side Validation:** Ensures user inputs are verified before reaching the server. This includes checking email formats, password strength, required fields, numeric inputs, and file formats for product images.
- **Responsive Design:** Navigation menus, product cards, images, and forms must automatically adjust to fit desktops, laptops, tablets, and smartphones seamlessly.
- **API Integration:** The frontend communicates dynamically with the backend using secure HTTPS RESTful API requests, updating the UI without requiring full page reloads.

Standard Folder Structure

```
/src
  /assets      # Static files (images, icons, styles)
  /components  # Reusable UI components (ProductCard, Navbar, etc.)
  /pages       # Main view structures (Home, ProductDetails, etc.)
  /services    # API integration files (axios/fetch setups)
  /store       # State management logic (Redux, Context API, etc.)
  /utils       # Helper functions (validation rules, formatters)
  App.js       # Root component
  index.js     # Entry point of the application
```

2. Development and Execution Process

Below is the standard workflow to develop, run, and build the component-based frontend application for ShopEZ:

Step 1: Prerequisites

Ensure you have Node.js and a package manager (like npm or yarn) installed on your computer.

Step 2: Install Dependencies

Navigate to your frontend project folder in your terminal and install the necessary libraries and packages defined in your package.json file.

```
cd shopez-frontend
npm install
```

Step 3: Environment Configuration

Set up your environment variables. You will typically need a `.env` file in your root directory to store your backend API endpoints securely.

```
REACT_APP_API_BASE_URL=http://localhost:5000/api
```

Step 4: Local Execution (Development Mode)

Start the development server. This will run the app locally and will feature hot-reloading so you can see changes instantly as you code.

```
npm start
```

Step 5: Building for Production

Once development is complete and the code is tested, you will compile the application into static files for deployment. This generates a `build` or `dist` folder containing optimized HTML, CSS, and JavaScript files ready to be hosted.

```
npm run build
```